Task 1: Load the Dataset

Develop a function to load the images from the dataset directory.

Your implementation should:

- Read images from all class folders.
- Resize every image to a common dimension.
- Convert images into NumPy arrays.
- Normalize the pixel values.
- Generate numerical labels for each class.
- Return the image dataset, labels, and class names.

Display:

- Total number of images
- Image dimensions
- Number of classes

Task 2: Split the Dataset

Split the dataset into training and testing sets using an appropriate train-test ratio.

Display:

- Training dataset size
- Testing dataset size

Task 3: Data Preprocessing

Perform the necessary preprocessing before training the model.

This may include:

- Reshaping the input data, if required.
- Feature scaling or normalization.
- Converting class labels into an appropriate format for multi-class classification.

Display the shape of the processed training and testing datasets.

Task 4: Design the Deep Neural Network

Design and implement a Deep Neural Network (DNN) suitable for image classification.

Your implementation should:

- Use the TensorFlow/Keras **Sequential** API.
- Include appropriate input, hidden, and output layers.
- Select suitable activation functions.
- Display the model summary.

Task 5: Compile the Model

Compile the model by selecting appropriate:

- Optimizer
- Loss function
- Performance metric(s)

Briefly justify your choice of loss function and output layer activation.

Task 6: Train the Model

Train the model using appropriate hyperparameters such as:

- Number of epochs
- Batch size
- Validation strategy

Store the training history for performance visualization.

Task 7: Evaluate the Model

Evaluate the trained model using the testing dataset.

Display:

- Test loss
- Test accuracy

Task 8: Generate Predictions

Use the trained model to predict the class labels for the testing dataset.

Compare the predicted labels with the actual labels.

Task 9: Performance Evaluation

Evaluate the classification performance by generating:

- Classification Report
- Confusion Matrix

Interpret the obtained results in a few sentences.

Task 10: Visualize the Training Process

Plot appropriate graphs to illustrate the learning behaviour of the model.

Include:

- Training and validation accuracy
- Training and validation loss

Each graph should contain:

- Title
- Axis labels
- Legend

Provide a brief interpretation of the plots.